

Context 1

The RoboCup@Home league

The RoboCup@Home league aims to develop robotic service and assistance technology with high relevance for future personal home applications. It is the largest annual international competition for autonomous service robots and is part of the RoboCup initiative. A set of benchmark tests is used to evaluate the robots' abilities and performance in a realistic, non-standard home environment. The focus is on the following domains, but is not limited to: Human-Robot Interaction and Cooperation, Navigation and Mapping in dynamic environments, Computer Vision and Object Recognition under natural light conditions, Object Manipulation, Adaptive Behaviors, Behavior Integration, Environmental Intelligence, Standardization and Systems Integration. It is co-located with the RoboCup symposium.

Who can participate?

Anyone with an independent robot can participate. The @Home league consists of a few tests and an open challenge to demonstrate your robot's skills. To participate in the open challenge it is necessary to participate in at least one test. The competition takes place in a real-world setting.

The competition

The RoboCup@Home competition consists of tests that robots must solve. The tests change over the years to become more advanced and act as a general measure of quality in the desired areas. Performance measurement is based on a score derived from competition rules and assessment by a jury.

Test criteria are listed below:

include human-robot interaction

be socially relevant

be application-directed/oriented

be scientifically challenging

be simple and have self-explanatory rules

be interesting to watch

The scenario

The final setting is the real world itself. To gradually develop possible technologies, a basic home environment is provided as a general setting. In the first years (we started in 2006) it consisted of a living room and a kitchen but soon it also began to involve other areas of daily life, such as a garden/park area, a store, a street or other public places.

Finals

RoboCup@Home ends with the finals where the teams with the highest points can play in the scenario that has been created. Teams are evaluated by a jury made up of roboticists and non-roboticists, such as people from industry, human-machine interaction, industrial design, the public and the press. The classification of the finals determines the winner. In the finals there is less focus on technical issues, as reaching the finals already means that the teams are very good at a technical level.

Context 2

Goiás

Goiás is a Brazilian state located in the Midwest region. Goiás borders the Federal District and the states of (from north clockwise) Tocantins, Bahia, Minas Gerais, Mato Grosso do Sul and Mato Grosso. The state capital is Goiânia. With 7.2 million inhabitants, Goiás is the most populous state in the Midwest and the 11th most populous in the country. It has the ninth largest economy among Brazilian federative units. In Brazil's geoeconomic division, Goiás belongs to the Centro-Sul (Center-South), being the northernmost state of the southern portion of Brazil. The state has 3.3% of the Brazilian population and is responsible for 2.7% of the Brazilian GDP.

The history of Goiás dates back to the beginning of the 18th century, with the arrival of pioneers from São Paulo. The

Rio Vermelho region was the first to be occupied, where Vila Boa (later renamed Goiás) was founded. The development and settlement of the state took place, in a more intensified way, starting in the 1930s with the founding of the new capital, Goiânia, and especially with the construction of Brasília in the 1950s.

Goiás is characterized by a landscape of chapadões (plateaus). In the height of the drought season, from June to September, the lack of rain makes the level of the Araguaia River go down and exposes almost 2 kilometres (1.2 mi) of beaches. At the Emas National Park in the municipality of Mineiros, it is possible to observe the typical fauna and flora from the region. At the Chapada dos Veadeiros the main attractions are the canyons, valleys, rapids and waterfalls. Other attractions include the historical city of Goiás and the hot springs of Caldas Novas.

History

The first European exploration of this interior part of Brazil was carried out by expeditions from São Paulo in the 17th century. Gold was discovered in the gravel of a tributary of the Araguaia River by the bandeirante Bartolomeu Bueno da Silva (the Anhanguera) in 1682. The settlement he founded there, called Santa Anna, became the colonial town of Goiás Velho, the former state capital. In 1744 the large inland area, much of it still unexplored by Europeans, was made a Captaincy General, and in 1822 it became a province of the empire of Brazil. It became a state in 1889. The Brazilian constitution of 1891 specified that the nation's capital should be moved to the Brazilian Highlands (Planalto Central), and in 1956 Goiás was selected as the site for the federal district and capital national, Brasília. The seat of the federal government was officially moved to Brasília in 1960.

Goiânia, the largest city and capital was planned in 1933 to replace the old, inaccessible former state capital of Goiás, 110 kilometres (68 mi) northwest. In 1937 the state government moved there, and in 1942 the official inauguration was held. Goiânia is now one of the fastest-growing cities in Brazil and one of the most livable cities in the country. It stands out as both an industrial center (along with the neighboring towns of Anápolis and Aparecida de Goiânia) and as a cultural center for country culture and music, known as Sertanejo

Due to the relatively large territory of the state, which was over 600,000 square kilometres (230,000 sq mi), communications were obviously very difficult. The northern part of the state began to feel abandoned by the southern government and began a movement for separation. Local political leaders also encouraged the movement. In 1988 the

northern half of Goiás became a separate state called Tocantins.

Economy

The service sector is the largest component of GDP at 43.9%, followed by the industrial sector at 35.4%. Agriculture represents 20.7%, of GDP (2004). Goiás exports: soybean 49.2%, meat of cattle 10.5%, gold 9.1%, other meat 7.5%, iron 7.4%, leather 4% (2002).

Goiás is a leader in the country in crop raising. In 2016, Goiás had the 3rd largest cattle herd in Brazil: 22.6 million head of cattle. The number of pigs in Goiás was approximately 2.0 million head in 2015. The State had the 6th largest Brazilian herd, 5% of the national herd. Among the municipalities in Goiás that stood out, Rio Verde had the 3rd largest national population. In 2016, Goiás was the 4th largest milk producer, accounting for 10.1% of the country's milk production. The number of chickens in the State was 64.2 million head in 2015. The production of chicken eggs this year was 188 million dozens. Goiás was the 9th largest producer of eggs, 5% of national production.